

A Quartic Predecessor Obstacle Problem

A concrete asymptotic question arising in causal zero-repair

August 2026

Abstract

We isolate a simple nonlinear predecessor operator on probability laws on $\mathbb{Z}_{\geq 0}$ arising from an exact causal zero-repair problem.

For the explicit pair

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right),$$

the probability generating functions satisfy the quartic cancellation

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}.$$

The predecessor operator sends a law B to the cheapest law A satisfying

$$Q * A \succeq_{\text{st}} P * B.$$

Iterating from δ_0 produces a rigorous sequence of feasible causal zero-repair roots, and therefore an explicit upper bound for the original finite-horizon problem.

Numerically, the corresponding mean inventory initially displays a striking approximately diffusive growth, roughly compatible with $c\sqrt{n}$ on moderate horizons, but longer computations exhibit pronounced active-set transitions. No asymptotic exponent is presently claimed.

The resulting question is a concrete infinite-dimensional piecewise-linear obstacle problem:

Determine the asymptotic growth of the predecessor iterates.

In particular, can their mean be bounded by $O(\log n)$, by $O(\sqrt{n})$, or must it grow faster?

1 The zero-repair problem

Let P, Q be the probability laws on $\{0, 1, 2, 3, 4\}$ defined above. They have the same mean and variance, while

$$G_P(z) - G_Q(z) = \frac{(1-z)^4}{64}. \tag{1}$$

A depth- n zero-repair tree is a family of probability laws

$$\{B_h\}_{|h| \leq n},$$

indexed by histories

$$h \in \{0, 1, 2, 3, 4\}^{\leq n},$$

such that at every internal node

$$Q * B_h = \sum_{x=0}^4 P(x) \delta_x * B_{hx}. \tag{ZR}$$

Let $\mathcal{K}_n$ denote the set of root laws admitting a depth- n tree, and define the minimum initial inventory

$$\boxed{\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} \mathbb{E}_B J.} \quad (2)$$

A separate argument gives the rigorous lower bound

$$\mathcal{B}_n = \Omega(\log n). \quad (3)$$

The matching upper bound

$$\mathcal{B}_n = O(\log n) \quad (4)$$

is open.

2 Stochastic upward closure

Write

$$A \succeq_{\text{st}} B$$

for first-order stochastic dominance.

The viability sets possess the following useful monotonicity property:

$$B \in \mathcal{K}_n, \quad \tilde{B} \succeq_{\text{st}} B \implies \tilde{B} \in \mathcal{K}_n. \quad (5)$$

This immediately gives the following extension principle.

Lemma 1 (Dominating predecessor). *Suppose*

$$B \in \mathcal{K}_n$$

and a probability law A satisfies

$$\boxed{Q * A \succeq_{\text{st}} P * B.} \quad (6)$$

Then

$$A \in \mathcal{K}_{n+1}.$$

Proof. By Strassen's theorem, couple

$$S \sim Q * A, \quad T \sim P * B$$

so that $S \geq T$ almost surely.

Write

$$T = X + J, \quad X \sim P, \quad J \sim B,$$

with X and J independent.

Conditional on $X = x$, define

$$B_x := \text{Law}(S - x \mid X = x).$$

Then

$$B_x \succeq_{\text{st}} B.$$

By stochastic upward closure,

$$B_x \in \mathcal{K}_n.$$

Moreover,

$$Q * A = \sum_x P(x) \delta_x * B_x.$$

Hence these B_x form the children of a valid one-step extension of A , and therefore $A \in \mathcal{K}_{n+1}$. $\square$

3 The homogeneous predecessor operator

Define

$$\boxed{D(B) \in \arg \min_A \{ \mathbb{E}_A J : Q * A \succeq_{\text{st}} P * B \}.}$$
 (7)

Start from

$$B_0 = \delta_0$$

and iterate

$$\boxed{B_{n+1} = D(B_n).}$$
 (8)

Let

$$m_n := \mathbb{E}_{B_n} J. \quad (9)$$

Lemma 1 yields inductively

$$\boxed{B_n \in \mathcal{K}_n, \quad \mathcal{B}_n \leq m_n.} \quad (10)$$

Thus the predecessor sequence is not merely a numerical heuristic: it is a genuine explicit upper-bound construction for the original causal problem.

The question is therefore reduced to understanding the growth of m_n .

4 Tail coordinates: an obstacle problem

For a law B define its tail sequence

$$b_k := \Pr_B(J \geq k), \quad k \geq 1,$$

and extend it by

$$b_k := 1, \quad k \leq 0.$$

Then

$$1 \geq b_1 \geq b_2 \geq \dots \geq 0 \quad (11)$$

and

$$\boxed{\mathbb{E}_B J = \sum_{k \geq 1} b_k.} \quad (12)$$

For a finitely supported probability law μ , introduce

$$(\mathcal{L}_\mu b)_k := \sum_i \mu(i) b_{k-i}. \quad (13)$$

This is precisely the tail sequence of $\mu * B$.

Consequently, if b is the tail sequence of B , the predecessor is the solution of the linear program

$$\boxed{\begin{array}{ll} \text{minimize} & \sum_{k \geq 1} a_k, \\ \text{subject to} & \mathcal{L}_Q a \geq \mathcal{L}_P b, \\ & 1 \geq a_1 \geq a_2 \geq \dots \geq 0. \end{array}} \quad (14)$$

Thus D is a linear optimization operator followed by projection onto the monotone probability-tail cone.

Now let S denote the backward shift

$$(Sb)_k = b_{k-1}.$$

Equation (1) gives the exact operator identity

$$\boxed{\mathcal{L}_P - \mathcal{L}_Q = \frac{1}{64}(I - S)^4.} \quad (15)$$

Writing

$$u := a - b,$$

the predecessor inequality becomes

$$\boxed{\mathcal{L}_Q u \geq \frac{1}{64}(I - S)^4 b,} \quad (16)$$

together with the obstacle constraints

$$1 \geq b_1 + u_1 \geq b_2 + u_2 \geq \cdots \geq 0. \quad (17)$$

This is perhaps the most compact formulation of the problem.

The forcing is a fourth discrete difference, while all positive cost is created by the requirement that the corrected sequence remain inside the monotone probability cone.

5 Numerical phenomenon

The iterates of (14) have been independently reproduced by two implementations. Some representative values are

n	m_n
5	0.504711...
10	0.730468...
20	1.06126...
40	1.48082...
80	2.02856...
100	2.2284806931361...
200	3.09531...

(18)

For example,

$$\begin{aligned} m_{10} - m_5 &\approx 0.225757, \\ m_{20} - m_{10} &\approx 0.330794, \\ m_{40} - m_{20} &\approx 0.419553, \\ m_{80} - m_{40} &\approx 0.547743, \\ m_{200} - m_{100} &\approx 0.866827. \end{aligned} \quad (19)$$

A useful diagnostic is

$$\alpha_n := \frac{\log(m_{2n}/m_n)}{\log 2}. \quad (20)$$

Numerically,

$n \rightarrow 2n$	α_n
$5 \rightarrow 10$	0.5334
$10 \rightarrow 20$	0.5389
$20 \rightarrow 40$	0.4806
$40 \rightarrow 80$	0.4541
$100 \rightarrow 200$	0.4742

(21)

On this window the data are strikingly compatible with diffusive scaling.
An even sharper diagnostic is

$$d_n := \frac{m_{2n} - m_n}{\sqrt{n}}. \quad (22)$$

For example,

$$d_{40} \approx 0.08661, \quad d_{100} \approx 0.08668. \quad (23)$$

If

$$m_n \sim c\sqrt{n} + b,$$

then

$$d_n \longrightarrow c(\sqrt{2} - 1),$$

and these values correspond to

$$c \approx 0.209. \quad (24)$$

However, this should *not* be read as evidence for a proved $\sqrt{n}$ law.

Longer numerical runs exhibit pronounced transitions in the active set of the linear program. The trajectory alternates between relatively smooth regimes and rapid “facet bursts.” Thus neither a logarithmic nor a square-root asymptotic law is currently justified by the numerical data alone.

6 Polyhedral interpretation

The map D is piecewise affine.

On any region where the same finite collection of inequalities in (14) is active, the next iterate is an affine function of the current tail profile.

The nonlinear behavior therefore occurs when the trajectory meets a new face of the monotone-tail cone or when a new stochastic-dominance constraint becomes active.

Schematically,

$$\boxed{\text{affine chart} \longrightarrow \text{facet collision} \longrightarrow \text{active-set change} \longrightarrow \text{new affine chart.}} \quad (25)$$

This suggests interpreting the growth of m_n as the accumulated cost of a discrete regulator which keeps a fourth-difference-driven linear evolution inside a positive monotone cone.

The analogy with reflected processes or obstacle problems is suggestive, but no limiting reflected diffusion is presently asserted.

7 The concrete open problem

Problem 2 (Quartic predecessor growth). *Let $B_0 = \delta_0$ and define*

$$B_{n+1} = D(B_n)$$

by (7).

Determine the asymptotic growth of

$$\boxed{m_n = \mathbb{E}_{B_n} J.}$$

In particular, decide which, if any, of the following estimates holds:

$$m_n = O(\log n), \quad (26)$$

$$m_n = O(\sqrt{n}), \quad (27)$$

or more generally

$$m_n = O(n^\alpha) \quad \text{for some } \alpha < \frac{1}{2}. \quad (28)$$

Several more structural versions may be useful.

Problem 3 (Active-set complexity). *Can one quantitatively control the number and total cost of active-set transitions of the LP (14) up to time n ?*

A bound of the form

$$\sum_{\substack{\text{facet transitions} \\ \text{up to } n}} \text{cost of transition} = O(\log n) \quad (29)$$

would be especially significant.

Problem 4 (Scaling limit). *After a suitable rescaling of space and mass, do the predecessor tails converge to a continuum obstacle problem or reflected evolution? If so, what is the correct scaling exponent?*

Problem 5 (Lyapunov functional). *Does there exist a functional $\mathcal{E}$ on monotone tails such that*

$$\mathcal{E}(\mathbf{D}(B)) - \mathcal{E}(B)$$

controls the increase in mean while telescoping over scales?

8 Why the answer matters

The predecessor construction is only a restricted subclass of all causal zero-repair trees, but it has one major advantage: it is automatically prefix-compatible.

By (10),

$$\mathcal{B}_n \leq m_n.$$

Consequently, a proof of

$$\boxed{m_n = O(\log n)} \quad (30)$$

would immediately imply

$$\mathcal{B}_n = O(\log n).$$

Together with the already established lower bound

$$\mathcal{B}_n = \Omega(\log n),$$

this would yield

$$\boxed{\mathcal{B}_n = \Theta(\log n)}. \quad (31)$$

On the other hand, if one could prove

$$m_n \asymp \sqrt{n}, \quad (32)$$

this would show that the homogeneous predecessor is a genuine but suboptimal causal mechanism and would sharply identify the additional multiscale structure required for the optimal logarithmic construction.

Thus either outcome is useful.

Community question. Given the obstacle recursion

$$\mathcal{L}_Q a \geq \mathcal{L}_P b, \quad 1 \geq a_1 \geq a_2 \geq \cdots \geq 0,$$

with

$$\mathcal{L}_P - \mathcal{L}_Q = \frac{1}{64}(I - S)^4,$$

and with a chosen to minimize $\sum_{k \geq 1} a_k$, what is the asymptotic growth of the iterates starting from $b^{(0)} = (0, 0, \dots)$?

Is there a useful connection to discrete obstacle problems, isotonic regression, reflected processes, free-boundary dynamics, polyhedral dynamical systems, or nonlinear Perron–Frobenius theory?